Discrete Linear Patterns

Concept

A discrete linear pattern changes by the same amount at each whole-number stage.

Common difference

Subtract consecutive terms to find d.

Term rule

$T = a + (n - 1)d$; a is the first term, n starts at 1.

Discrete graph

Plot separate points when half-stages have no meaning.

Membership

Solve for n; accept only allowed whole-number stages.

Example 1

Find a rule and term 20 for $7, 11, 15, 19, \ldots$.

Solution

1. $d = 4$, so $T = 7 + 4(n - 1) = 4n + 3$.

2. At $n = 20$: $T = 4(20) + 3 = 83$.

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Example 2

Is 100 a term of $T = 4n + 3$?

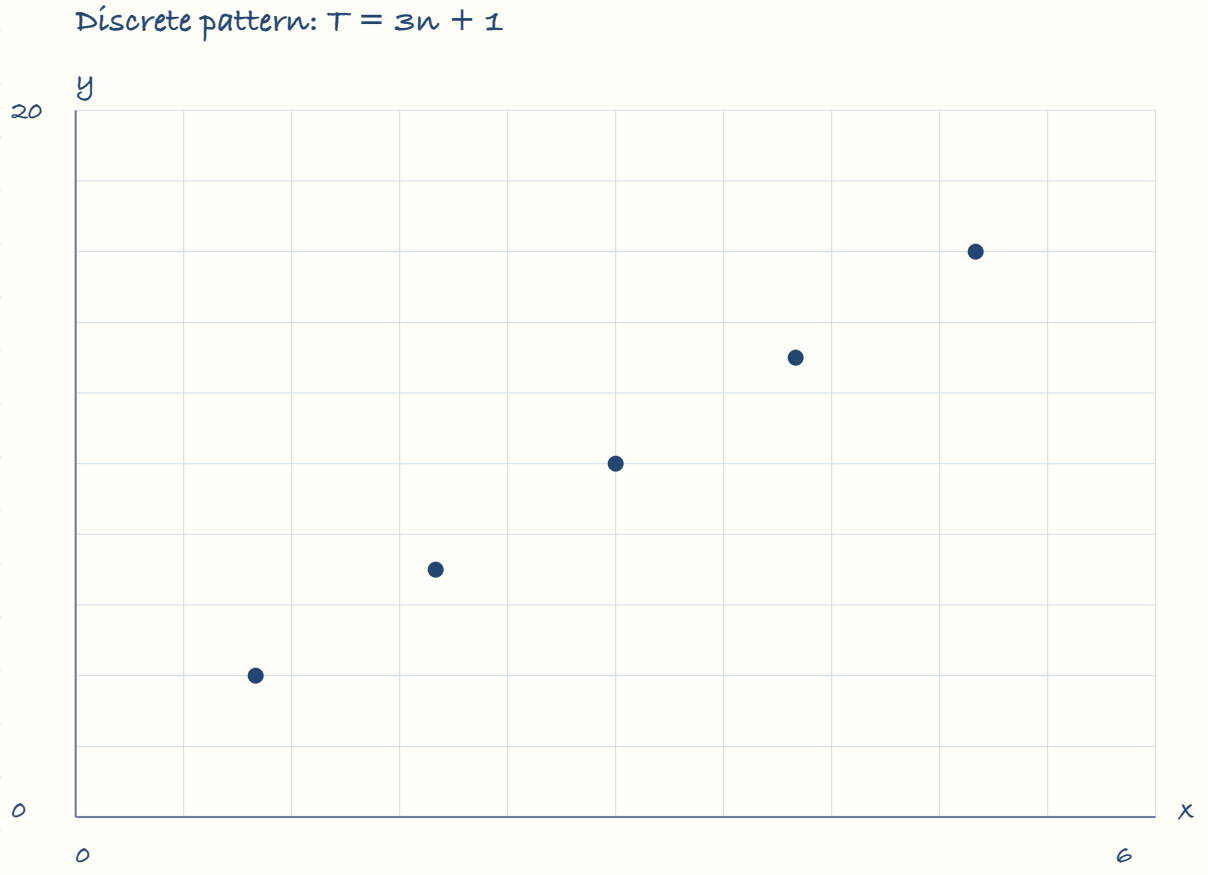
Solution

1. $100 = 4n + 3 \rightarrow n = 97/4 = 24.25$.

2. No: the stage number is not a whole number.

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Visual notes | read the graph



Each dot is a whole-number stage. There is no half-stage in this model.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Write a rule for 5, 8, 11, 14, … .

2. Find its 30th term.

3. For $T = 6n + 2$, list the first four terms.

4. Is 74 a term of $T = 6n + 2$?

5. Explain why $y = n^2$ does not describe a linear pattern.

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Practice answers | check after trying

1. $T = 3n + 2$.

2. $3(30) + 2 = 92$.

3. 8, 14, 20, 26.

4. $6n + 2 = 74$ gives $n = 12$, so yes.

5. Outputs 1, 4, 9, 16 have differences 3, 5, 7. The difference is not constant.